# Reading the Market: News-Augmented Regime Discovery with Unsupervised Representation Learning

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## 1. Goals and Motivation

This project investigates whether incorporating news textual information into market regime detection yields regimes that are more stable, more interpretable, or more economically meaningful than those obtained from price-based features alone. Traditional regime discovery methods — HMMs, GMMs, and threshold-based switching models — rely almost exclusively on price-derived features, which by construction only reflect information after it has already been incorporated into market prices. Market news provides a direct view of market narratives, investor attention, and information flow. If text-based signals help identify meaningful regimes, this project would show that financial news can serve not only as a sentiment or prediction input, but also as an unsupervised representation of changing market states.

## 2. Key Related Work

This project builds on existing work in regime discovery for financial time series, news-based measurement of macroeconomic and market conditions, classical and neural topic modeling for financial text, and domain-specific language models for finance. Specific references are listed at the end of this proposal.

## 3. Data and Codebases

The project uses two main data sources. News data comes from RavenPack financial headlines spanning 2000–2025, covering multiple well-documented market regimes; market data includes daily price, volatility, sector returns, and selected macro-financial variables for major US equity indices from WRDS. Implementation will rely on standard Python libraries for clustering, latent-state, and classical topic models, together with HuggingFace `transformers`, `sentence-transformers`, and BERTopic for text representation, with FinBERT as the primary domain-specific model. PyTorch may be added if neural representation components are incorporated.

The components I will implement myself include the end-to-end data pipeline aligning headlines with market data, the fusion strategies combining price- and text-based features, the evaluation modules covering clustering quality, regime stability, and financial interpretability, and a regime characterization module that aggregates news within each discovered regime to produce semantic summaries.

## 4. Planned Experiments

The experimental plan is organized around three method pathways. The price-only baseline applies clustering and latent-state models to standard market features; the news-only pathway applies the same model family to time-aggregated news representations; and the hybrid pathway combines the two through fusion strategies ranging from concatenation to more structured joint representations.

Evaluation covers four dimensions: *statistical fit*, *clustering quality*, *regime stability* under bootstrap resamples and rolling windows, and *financial interpretability*, examined both quantitatively through per-regime financial quantities and qualitatively through semantic characterization of regimes against known historical episodes.

## 5. Computing Resources

Most of the project can run on a personal laptop. The most computationally expensive step is embedding extraction over the full RavenPack corpus using FinBERT or sentence-transformer models, for which I plan to use Google Colab. The overall compute footprint is expected to be modest.

## 6. Projected Timeline

| Week | Milestone |
|---|---|
| **Week 1 (Apr 30 – May 6)** | Data ingestion, cleaning, temporal alignment of RavenPack and market data; exploratory analysis; finalize feature selection. |
| **Week 2 (May 7 – May 13)** | Construct market, text, and hybrid feature sets (sentiment, LDA, BERTopic, FinBERT embeddings); fit and validate price-only baseline (HMM, GMM). |
| **Week 3 (May 14 – May 20)** | Run regime detection on text-only and hybrid inputs; full four-dimensional evaluation (statistical fit, clustering quality, stability, interpretability). |
| **Week 4 (May 21 – May 27)** | Synthesize results, qualitative regime characterization via news aggregation, final report and presentation slides. |

## References


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